

Reviewer 2 — Response & Correction Plan

IEEE Access manuscript: "An Interpretable Deterministic Framework for Cross-Language Educational Code Similarity Using Code Property Graphs"

This document turns Reviewer 2's comments into a concrete task list. For each comment: what the reviewer said, why it's a real issue, what to fix, and where in the manuscript to fix it. Work through the items in order — items 1 and 4 are outright numerical errors and should be corrected first; items 2, 3, 5, and 6 require new analysis or rewriting.

Overview

#	Issue	Type	Owner action
1	Table 10 aggregate doesn't match exclusion claim	Arithmetic error	Recompute; pick one consistent aggregate; fix Abstract/Contributions/Conclusion
2	Zero-CES rate drops 23.25% → 0.25% after filtering	Selection bias, undiscussed	Report unfiltered baseline; add validity-threat note
3	No single-view (BL/WL/CES-only) ablation	Missing analysis	Compute and tabulate 3 isolated-view runs
4	64% adversarial floor mixes n=93 and n=60	Computation error	Recompute from the correct n=60 IRR sample
5	TF-IDF baseline is really an internal ablation	Framing issue	Relabel; add a true external IR baseline
6	Phase 1 details (weights, 9-pattern taxonomy) underdescribed	Self-containment gap	Expand Section III-A with full derivation

1. Table 10 aggregate arithmetic

Reviewer's comment:

“Table 10's cells sum to 349/400 (87.25%), which means that the n=11 and n=3 cases are included in the aggregate. Please either report the corrected aggregate (342/386 = 88.60%) or withdraw the exclusion claim.”

Why this is a real problem:

The text around Table 10 states that the C → Java (n=11) and C++ → Java (n=3) directions are “statistically insufficient for directional inference and are excluded from all aggregate accuracy claims.” But $140+246+11+3 = 400$, and $128+214+6+1 = 349$ — the reported 87.25% aggregate already includes those two underpowered directions. The exclusion claim and the reported number contradict each other.

What to do (pick one path):

1. Relabel 87.25% as the full-corpus aggregate (all 400 pairs, all four directions) and delete the sentence claiming exclusion.
2. Add a second, explicitly separate “core-direction aggregate” computed only over Java → C/C++ and C ↔ C++ (342/386 = 88.60%), and present this as the headline number given the power concerns already raised in Section VI-B / VII-E.

Recommended: do both — report the full aggregate (87.25%) and the core-direction aggregate (88.60%) side by side, with the exclusion logic stated correctly for the latter only.

Where to fix it:

- Table 10 caption and surrounding paragraph (Section VI-B)
- Abstract (currently states 87.25% with an exclusion claim)
- Contributions list in Section I (“Weight Stability Against Cross-Language Testing” bullet)
- Conclusion (Section VIII)

2. Selection bias from corpus filtering

Reviewer's comment:

“Corpus filtering reduces the zero-CES rate from 23.25% to 0.25%, introducing a potential selection bias that is not discussed in the validity analysis. Please report baseline results on the unfiltered corpus.”

Why this is a real problem:

Table 16 shows the ablation (uncurated) C++ dataset has a 23.25% zero-CES failure rate, but the curated 400-program cross-language corpus has only 0.25%. The curated corpus was built by “selecting compilable and interpretable programs with entry point identification,” which likely removes exactly the hard cases the framework struggles with. This is not currently named as a validity threat.

What to do:

3. Add a supplementary table reporting top-1 accuracy on the unfiltered/uncurated pool (before compilability and entry-point filtering).
4. Include the zero-CES fallback-to-WL accuracy (90.3%, already reported in Section VI-E) alongside it so the reader sees the realistic floor.
5. Move the current filtering criteria into an explicit “Corpus Construction” subsection in Section V-A.
6. Add a named “selection bias” entry to the Threats to Validity section (Threat Group 1 or 2), distinct from the existing “coverage” framing.

Where to fix it:

- Section V-A (Dataset Construction)
- Section VI-E (CES Extraction Failure Analysis) — add the unfiltered comparison table here or reference it
- Section VII-E (Threats to Validity), Threat Group 1 or 2

3. Missing single-view ablation

Reviewer's comment:

“No single-view ablation is provided. Given Table 8's optimum weights (0.00, 0.05, 0.95), the contribution of the structural view remains unclear. Please report BL-only, WL-only, and CES-only results.”

Why this is a real problem:

Table 8 shows that the in-sample C++ optimum weights are ($w_{BL}=0.00$, $w_{WL}=0.05$, $w_{CES}=0.95$) — almost all weight on CES, almost none on WL. This raises a legitimate question the paper doesn't answer: is the WL (structural) view actually contributing anything, or is CES doing nearly all the work?

What to do:

7. Compute top-1 accuracy on the 400-program cross-language corpus with weights set to isolate each view: (1,0,0) for BL-only, (0,1,0) for WL-only, (0,0,1) for CES-only.
8. No new extraction is needed — each view's score is already computed for late fusion (Eq. 3); this just requires isolating and reporting each term instead of only the fused score.
9. Add a new ablation table (near Table 8 or Table 12) with columns: View, Weight config, Top-1 Acc., 95% Wilson CI.
10. In the text, state explicitly whether the fused model's advantage over TF-IDF (Table 12) comes mainly from CES, mainly from WL, or genuinely from combining both.

Where to fix it:

- New table near Table 8 (Section IV-C / VI-A) or Table 12 (Section VI-B)
- Discussion in Section VII-A or a new short subsection

4. The 64% adversarial floor computation

Reviewer's comment:

“The reported 64% adversarial floor combines incompatible sample sizes ($n=93$ and $n=60$). Please recompute.”

Why this is a real problem:

In Section VII-E, Threat Group 1, the manuscript computes $(349-93)/400 = 64.0\%$ as a “lower bound” based on submissions where raters disagreed. But $n=93$ is actually the zero-CES ablation count from Table 16 (the uncurated C++ ablation set) — it has nothing to do with the $n=60$ stratified sample used for the IRR (Cohen's κ) study described in Section V-B and referenced in Section VII-E. This looks like a copy-paste or variable-naming mistake, and the resulting 64.0% figure is not meaningful as written.

What to do:

11. Identify the actual number of rater disagreements within the $n=60$ stratified IRR sample (this is recoverable from $\kappa=0.86$ and the raw agreement/disagreement counts, which should already exist in the annotation records).
12. Report the $n=60$ sub-sample accuracy under two conventions: (a) using majority/lead-author label, and (b) treating every disagreement as an automatic failure — rather than extrapolating a rate to the full $n=400$ set.
13. Explicitly withdraw the current 64.0% figure and replace it with the corrected number and method.

Where to fix it:

- Section VII-E, Threat Group 1 (Dataset Scope and Annotation Reliability)

5. TF-IDF as internal ablation vs. external baseline

Reviewer's comment:

“TF-IDF is an internal component of the framework, making the McNemar comparison an ablation rather than an independent baseline. Please clarify this distinction or compare against an external IR baseline.”

Why this is a real problem:

The “TF-IDF educational baseline” in Table 12 is architecturally the same Baseline (BL) view already used inside the fusion model (Eq. 3) — it is the framework run with weights (1,0,0) rather than an independently designed IR system. Listing it in Table 2/Table 12 next to MOSS, JPlag, and the neural models overstates the McNemar significance test as evidence of beating independent prior art, when it is really an ablation result.

What to do:

14. Relabel this row in Table 2 / Table 12 explicitly as an ablation baseline, separated visually from the truly external methods (MOSS, JPlag, CodeBERT, GraphCodeBERT, UniXcoder).
15. Add a genuinely external IR baseline — e.g. classic LSA-based code retrieval (Cosma & Joy, ref [14], already cited) or a plain bag-of-identifiers cosine baseline without any of the framework's normalization/weighting machinery.
16. Keep the existing TF-IDF-vs-framework McNemar test, but reframe it in text (Section VI-B / VII-B) as quantifying the incremental contribution of WL+CES over the BL view alone — this connects naturally to the new ablation table from item 3.

Where to fix it:

- Table 2 (Section II-E) and Table 12 (Section VI-B)
- Section VI-B (Statistical analysis paragraph) and Section VII-B

6. Phase 1 details underdescribed

Reviewer's comment:

“Phase 1 provides the fusion weights, taxonomy, and 88.5% baseline but is neither cited nor sufficiently described. Please provide enough detail for this paper to stand on its own.”

Why this is a real problem:

Several load-bearing details — the fusion weights (w_{BL} , w_{WL} , w_{CES}) = (0.35, 0.40, 0.25), the origin of the 9 baseline CES patterns that Table 4 extends, and the 88.5% Phase 1 baseline — are used throughout this paper but only briefly summarized in Section III-A, with the reader referred to a Phase 1 manuscript that is itself still under review and therefore cannot be formally cited yet. This makes it hard to evaluate this paper independently.

What to do:

17. Expand Section III-A into a fuller, self-contained description: the 104-program C dataset, the grid-search procedure used to fit the fusion weights, and the WL kernel configuration ($h=2$) as originally validated.
18. Explicitly list and define the 9 baseline CES patterns (currently only implied by the “Ph. 1 & 2” column in Table 4) so Table 4 doesn't require external context to interpret.
19. Cite Phase 1 only as “manuscript in preparation” for provenance/attribution, not as the source of load-bearing methodological detail needed to understand this paper.

Where to fix it:

- Section III-A (Phase 1 Architecture and CPG Unification) — primary expansion point
- Table 4 caption / Section III-C (Extended CES Taxonomy)
- References list — add or correct the Phase 1 citation format

Before resubmission — final checklist

- Table 10, Abstract, Contributions (Section I), and Conclusion (Section VIII) all report the same, correctly-labeled aggregate figure(s).
- Unfiltered-corpus baseline table added, with a matching validity-threat entry.
- New single-view ablation table (BL-only, WL-only, CES-only) added and discussed.
- 64% adversarial floor recomputed from the correct $n=60$ sample and clearly explained.
- TF-IDF row relabeled as an ablation; a true external IR baseline added.
- Section III-A expanded so the paper is self-contained without relying on the Phase 1 manuscript for essential detail.
- Re-read the Abstract and Conclusion last, since items 1 and 6 both touch numbers/claims repeated there.